

**Land-use History Shown to Affect the Impact of Control Burns on Soil
Fungi Abundance Without Shifting Broad Functional Roles**

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Abstract

One method used during land conservation efforts is controlled burns, which act to remove leaf litter while supplying the soil with nutrients. Studies have shown the effectiveness of control burns on the surrounding plant life and fungi. But not much research has considered the history of the land which is being conserved. We attempt to fix this gap by looking at the effectiveness of control burns on differing land-uses: post-agricultural or remnant on the overall fungi abundance and function in the soil that was burned. By collecting ecological data on the fungi found at different plots in South Carolina's Longleaf Pine Savanna, we were able to categorize the fungi as beneficial or pathogenic using FUNGuildR. We ran PERMANOVA tests and Non-Metric Multidimensional Scaling to find any significance between land-use histories impact on the control burns effect on soil fungi abundance and function. Our results show that control burns have more impact on post-agricultural lands soil fungi abundance than on remnant land. In both land-uses the control burns had no impact on overall fungi soil function.

1. Introduction

Controlled burns are frequently used to manage and restore plant life in fields across the southern United States. We collected fungal community data from within the longleaf pine savanna ecosystem (*Pinus palustris*) near the border of South Carolina. Longleaf pine savanna is a species-rich system that was historically maintained by frequent, low-intensity fires that

created an open canopy of pines with a diverse herbaceous understory. More than 97% of this habitat type has been lost and there are efforts to restore the habitat ongoing (Noss 2012).

Understanding the impacts of agricultural land use and fire regimes is critical for conserving and restoring this ecosystem. Many other studies were recently done looking at exactly these issues in this environment (Turley, Nash E., et al. 2020). In our study, we try to research the question of whether controlled burns contribute to increasing beneficial or pathogenic fungi regarding both post-agricultural and remnant land-use. Soil fungi is an important part of the ecosystem. Fungi's ability to decompose organic matter helps put nutrients into the soil which benefit the surrounding plants. Therefore, studies regarding soil fungi are important for understanding what conservation methods may be best. Recent research at the same location has evidence that fire frequency is a primary driver of the microbial community structure (Link, Arielle C., et al., 2026). We wish to research if the land-use history of each plot influences the fire frequency as a driver. Our hypothesis is that an overall significant and stronger positive relationship will occur between the number of controlled burns and abundance of beneficial fungi for post-agricultural land compared to remnant land.

II. Methods

Our data was collected at the Savannah River Site (SRS), a National Environmental Research Park near Aiken, SC. This site has restored longleaf pine savanna habitat with two types of land use history (post-agricultural and remnant). Post-agricultural land use corresponds to farmland that was last in agriculture in 1950 and has not been farmed since that

time. Remnant land use corresponds to land that has never experienced agriculture. The exact locations used in this study vary in fire frequency, with some sites experiencing relatively few fires and some sites experiencing relatively more fires, with a range of 1-12 fires since 1991. To examine the effects of land use, half of our sites were post-agricultural and half were remnant (Brudvig et al., 2014). We collected soil to sample the fungal community in June 2022. We collected approximately 0.1 kg of soil from the top 2 cm of the soil. We kept these samples in sterile conditions and frozen at -40°C until they were analyzed. Samples were processed through the Microbial Community Sequencing Service Center at the University of Colorado to obtain fungal internal transcribed spacer (ITS) sequencing data (Suleiman et al., 2019). ITS sequence data were divided into operational taxonomic units (OTUs) by similarity, with 5% difference being the defining difference between OTUs. Sequences were run through GenBank to assign taxonomic identity to the finest resolution possible. To analyze the data, we inputted our data into RStudio to use both the vegan packages for testing and visualization, tidyverse for data cleanup and organization, and FUNGuildR for functional trait assignment. Data was first run through FUNGuildR package's funguild_assign function to assign identifiable OTUs as a combination of Saprotroph, Symbiotroph, and/or Pathotroph (Nguyen NH et al. 2016). We used these assignments to categorize the OTUs as beneficial, pathogenic, or neutral. All assignments that were strictly Symbiotrophs were placed as beneficial, Pathotrophs as pathogenic, and anything else was labeled neutral. To perform statistical tests and graphing we used the independent variables land-use (post-agricultural or remnant), categories (beneficial or pathogenic), and fire frequency (1-12) to determine any significant change in dependent variables of the overall fungi function and diversity. Vegan's adonis2 function was used to

perform a Permutational Multivariate Analysis of Variance (PERMANOVA) statistical test to evaluate the effect of land-use history and fire frequency on fungi function and overall abundance. Significance was determined using 999 permutations on our test. For visualization, the vegan package was used to produce two Non-Metric Multidimensional Scaling (NMDS) plots using the Bray-Curtis dissimilarity matrix. This visualization technique allowed us to project our multi-variable data into a two-dimensional graph ($k = 2$). The variables of fire frequency and land-use categories were fitted onto the NMDS plot using vegan's `envfit` function. This allowed vector fitting for the fire frequency and factor centroids for the land-use to be included on the final plot. Further simple linear regression statistical tests were done to determine the significance of the relationship between fire frequency and beneficial or pathogenic abundance.

III. Results

NMDS was used to determine the functional change of the overall fungi in the soil from the fire frequency of each land-use. The NMDS ordination ($k = 2$ with a stress of 0.031) has the resulting graph showing a high overlapping between remnant and post-agricultural land-use histories (Figure 1). Fire frequency was included on the NMDS graph via vector fitting (`envfit`) and yielded a P-value of 0.33 and $R^2 = 0.0689$. Permutational multivariate analysis of variance (PERMANOVA) further reports that land use history ($P = 0.827$) nor fire frequency ($P = 0.324$) have a significant effect on the function of fungi community. The proportion of variance was also shown to be low in both fire frequency ($R^2 = 0.0689$) and land use ($R^2 = 0.0057$). Overall PERMANOVA test had an overall $P = 0.738$ and $R^2 = 0.4621$.

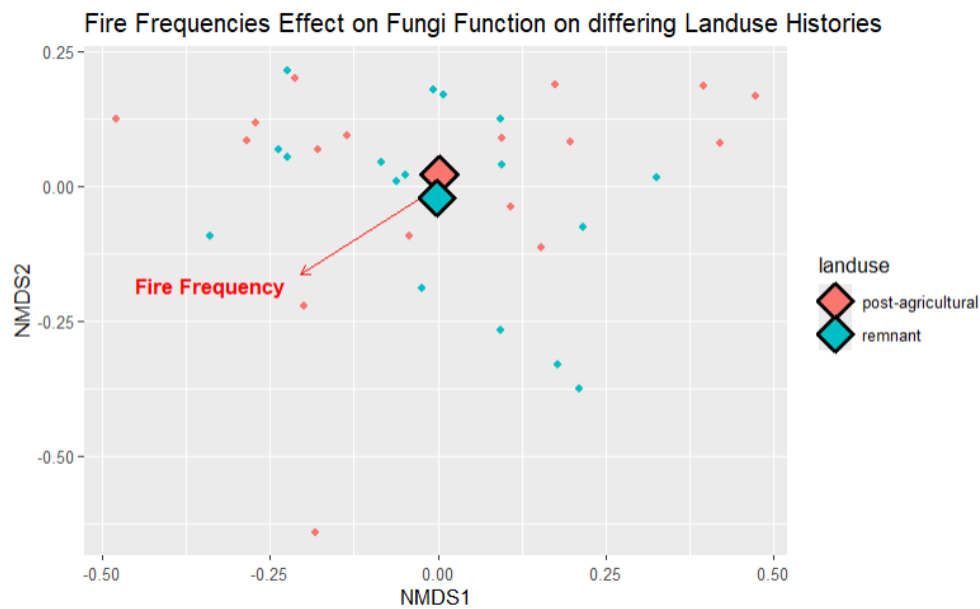


Fig 1. NMDS showing relationship between beneficial fungi count and burn frequency across different land-use histories (remnant or post-agricultural). Points show significant overlapping and no direct relationships. ($P = 0.738$ and $R^2 = 0.4621$). Centroids display the center of each land-use distribution, which nearly completely overlap with one another. The red arrow indicated that the direction in which fire frequency increases aims at the bottom-left of the graph. This was accomplished via vector fitting (P -value = 0.33 and $R^2 = 0.0689$).

Further analysis was done with simple linear regression for both beneficial and pathogenic fungi abundance relationship with total fires per plot. The test between beneficial fungi and fire frequency yielded a insignificance result with a P -value of 0.2275 and R^2 value of 0.01534 .

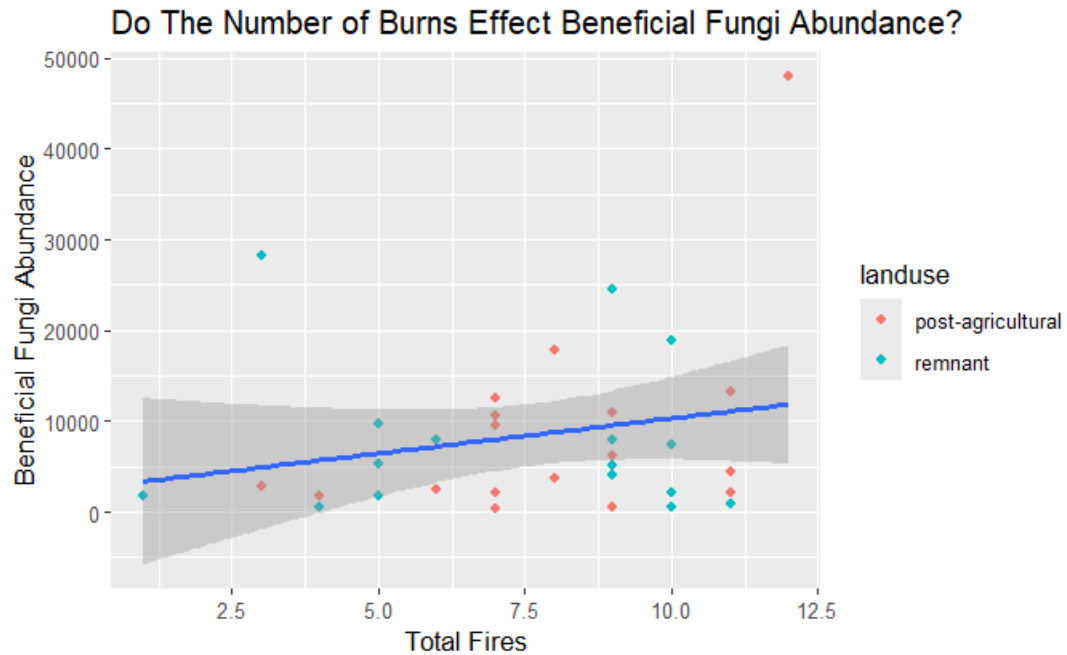


Fig. 2. Shows relationship between beneficial fungi count and total burns at each plot across both land-use histories (remnant or post-agricultural). The overall trend line shows a slight positive relationship in beneficial fungi abundance and total fire, but the linear regression model gives a high P value ($P = 0.2275$) and low R^2 value ($R^2 = 0.01534$).

The simple linear regression between pathogenic fungi and fire frequency also showed low significance with a P-value of 0.223 and R^2 of 0.046.

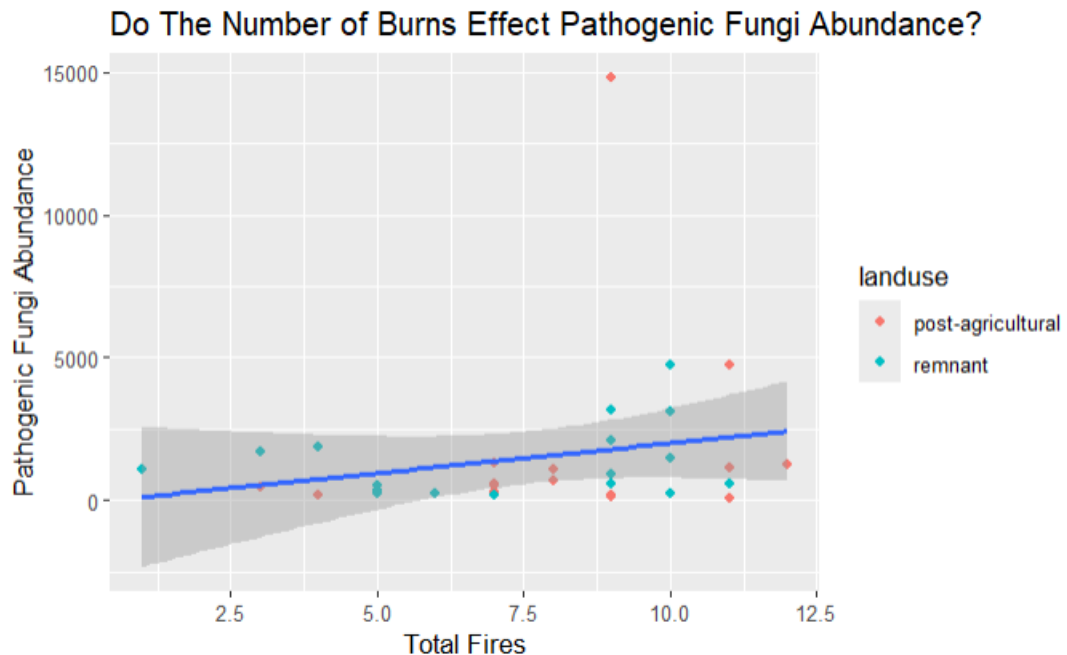


Fig. 3. Shows relationship between pathogenic fungi count and total burns at each plot across both land-use histories (remnant or post-agricultural). The overall trend line shows a slight positive relationship in pathogenic fungi abundance and total fire, but the linear regression model results in a high P value ($P = 0.223$) and low R^2 value ($R^2 = 0.046$).

NMDS was also run to determine the overall fungi abundance change from the fire frequency for each land-use. This test was ran using the uncategorized OUT data. The NMDS ordinance ($k = 2$ with a stress of 0.21) has the resulting graph showing distinct but overlapping ellipses (Figure 4). Fire frequency was included on the NMDS graph via vector fitting (envfit) and yielded a P-value of 0.034 and $R^2 = 0.1983$. Notably, the post-agricultural ellipsis follows the fire frequency vector most closely than the remnant ellipses. PERMANOVA further reports that the overall diversity of the fungi community had significantly been altered by the fire ($P = 0.004$) with the proportion of variance was also shown to be low ($R^2 = 0.09504$).

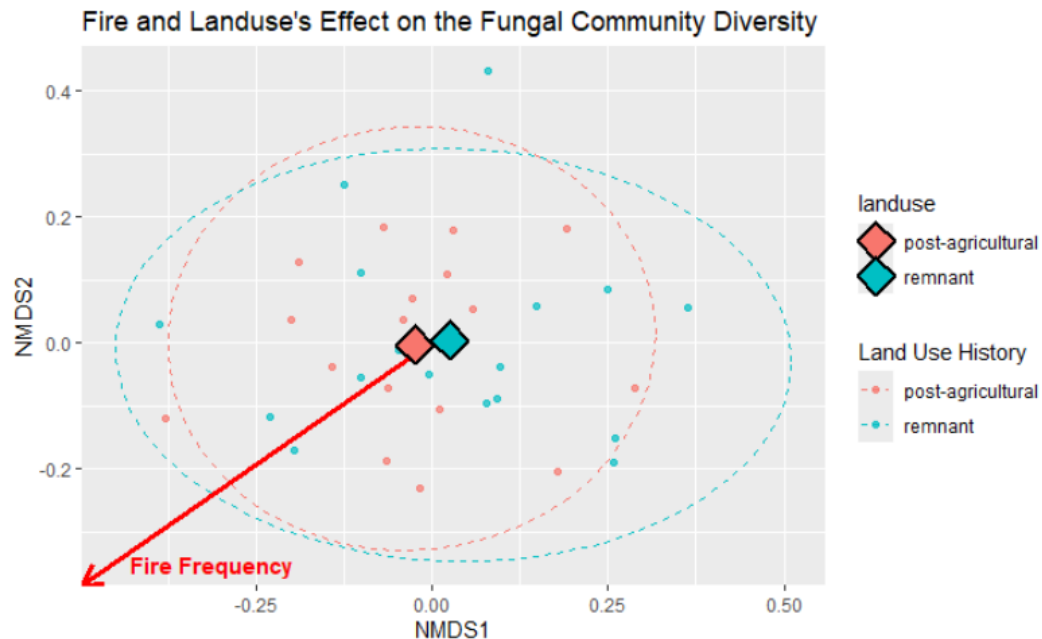


Fig. 4. NMDS showing relationship of overall fungi abundance and burn frequency across different land-use histories (remnant or post-agricultural). Points show overlapping but distinct ellipses ($P = 0.004$ and $R^2 = 0.09504$). The red arrow indicated that the direction in which fire frequency increases aims at the bottom-left of the graph. The fire frequency arrow was accomplished via vector fitting ($P\text{-value} = 0.034$ and $R^2 = 0.1983$).

IV. Discussion

Our tests conclude that the use of control burns has minimal impact on the overall fungi community function, but significant impact on the overall fungi diversity. All p -values associated with fungi function were insignificant (> 0.05) which heavily infer no significant relationship between the frequency of controlled burns, beneficial and pathogenic fungi abundance, and

land-use history. This directly goes against our original hypothesis which we predicted that post-agricultural land would be a more significant driver of increased beneficial fungi compared to remnant land. Further linear regression analyses looked at the relationships between beneficial or pathogenic abundance and fire frequency separately and still resulted in no significant relation. Our result mimics recent studies looking at fire frequency effect on the fungal community (Cuberos et al. 2025, Oliver et al. 2015) which suggest a burn management has a minimally distinguishable effect on long-term fungal function. These test results imply that the use of control burns may not have as significant an effect on the overall fungi community than previously thought. However, when testing without categorizing the function of the OTUs as beneficial or pathogenic, we see a significant relationship between fire frequency, land-use history, and change within the overall fungal community diversity. Our NMDS has a notable difference in ellipses of the land-uses, where post-agricultural is following the direction of the fire frequency more closely than the remnant. This suggests that although the fungi function was not impacted, the actual diversity of the fungi in the soil was. Moreover, the control burns impacted the post-agricultural plots fungi diversity more than the remnant plots.

We could possibly explain the results by pointing to the many differences between each fungus. Even if fire were to remove a beneficial fungi OTU, it is possible it could also allow another beneficial OTU to thrive and take its place; resulting in an insignificant gain or loss for the overall function but a significant change in the diversity of the overall fungi.

The results of our study could have been impacted by potential shortcomings. Our three classifications (beneficial, pathogenic, and neutral) could have been too broad to properly label

each fungi OTU found on the plots. If we oversimplified the fungi in the study, any evidence of effect could have been accidentally removed. Furthermore, OTUs species are notoriously known to be hard to identify. There is a possibility that FUNGuildR could have mislabeled or failed to identify some OTUs when we used it to determine if the OTUs found were beneficial, pathogenic, or neutral.

The importance of maintaining environments, such as the Longleaf Pine Savannah, relies partly on understanding the soil's fungi community and what methods are effective at removing pathogenic fungi and increasing beneficial fungi. Future studies may be inclined to look further into the methods effective across all land-use types, proving a significant positive relationship between control burns and overall fungi community, and better methods of classifying fungi.

Works Cited

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